

Q1: A pressure control system includes a thin steel disk that is to close an electrical circuit by deflecting 1 mm at the center when the pressure attains a value of 3 MPa. Calculate the required disk thickness if it has a radius of 0.030 m and is built-in at the edge. Let $\nu = 0.3$ and $E = 200$ GPa.

Solution

We consider the disk as uniformly loaded solid circular plate and built-in at the edges. The deflection equation obtained in the notes for this model is (Eq.(1))

$$w = A_0 r^2 \ln r + B_0 \ln r + C_0 r^2 + D_0 + \frac{P_0 r^4}{64D} \quad (1)$$

that

$$D = \frac{Et^3}{12(1 - \nu^2)} \quad (2)$$

For solid plates the deflection and slope at the center of the sheet must be limited

$$\lim_{r \rightarrow 0} w = \text{finite} \quad (3)$$

So in the term $\left(\frac{B_0}{r}\right)$, B_0 must be 0. Besides that, both M_r and Q_r must be finite.

$$\lim_{r \rightarrow 0} M_r = \text{finite} \quad (4)$$

$$\lim_{r \rightarrow 0} Q_r = \text{finite} \quad (5)$$

Which, in Eq.(5), A_0 must be zero. The deflection equation simplifies as follows

$$w = C_0 r^2 + D_0 + \frac{P_0 r^4}{64D} \quad (6)$$

For built in BC, we have

$$\text{@ } r = 0.03 = a \quad \rightarrow \quad w = 0, \quad w_r = 0 \quad (7)$$

$$\lim_{r \rightarrow a} w = 0 \quad (8)$$

$$\lim_{r \rightarrow a} M_r = 0$$

$$\lim_{r \rightarrow a} Q_r = 0$$

$$\lim_{r \rightarrow a} w = 0 \quad (9)$$

At least we obtain Eq.(9). In question is defined the maximum deflection is 1 mm, and it is in the middle of the disk ($r = 0$). So,

$$\boxed{\dots} \quad (10)$$

Substituting D from Eq.(2)

$$\boxed{\dots}$$

$$\boxed{\dots}$$

$$\boxed{t = 1.275 \text{ mm}}$$

Finally, we should check the ratio of maximum deflection to thickness which should be fewer than 0.1:

$$\boxed{\dots}$$

This result does not satisfy the thin plate assumption.

Q2: Consider a circular plate of radius a and flexural rigidity D , subjected to a rotationally symmetric lateral load that increases linearly from the center to the edge according to the expression:

$$p(r) = p_0 \frac{r}{a}$$

where p_0 is the load intensity at the boundary ($r = a$), and r is the radial distance from the center.

- a) Assuming the plate is simply supported along its edge ($r = a$), derive the governing differential equation and show that the resulting lateral displacement $w(r)$ is given by:

$$w = \frac{p_0 r^5}{225aD} + c_3 r^2 + c_4$$

where the constants of integration are:

$$c_3 = -\frac{p_0 a^2}{90D} \left(\frac{4 + \nu}{1 + \nu} \right), \quad c_4 = \frac{p_0 a^4}{45D} \left[\frac{4 + \nu}{2(1 + \nu)} - \frac{1}{5} \right]$$

and ν represents Poisson's ratio.

- b) Resolve the problem by changing the boundary condition to a clamped edge at $r = a$. Determine the complete analytical expression for the deflection $w(r)$ under this condition.
- c) Obtain and compare the radial stress $\sigma_r(r)$ in both the clamped and simply supported plates across the radius, highlighting the behavior at the center ($r = 0$) and at the boundary ($r = a$).
- d) Investigate if it is possible to draw a normalized, non-dimensional figure for this specific loading ($p = p_0 \frac{r}{a}$) that is conceptually similar to Figure 4.4 of your textbook.

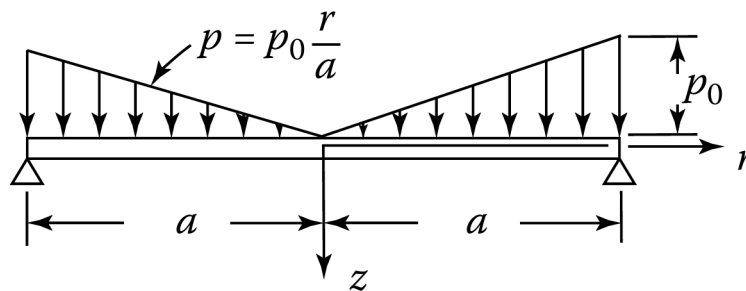


Figure 1: Circular plate under linearly varying load

Solution

We consider the disk as a solid circular plate under an axisymmetric load $p(r) = p_0 r/a$. The governing equation is

$$\nabla^4 w = \frac{p(r)}{D}, \quad \nabla^4 w = \frac{1}{r} \frac{d}{dr} \left\{ r \frac{d}{dr} \left[\frac{1}{r} \frac{d}{dr} \left(r \frac{dw}{dr} \right) \right] \right\} = \frac{p_0 r}{aD} \quad (11)$$

The general solution is

$$w = A_0 r^2 \ln r + B_0 \ln r + C_0 r^2 + D_0 + w_p \quad (12)$$

For a solid plate, deflection and slope at the center must remain finite, so from the slope expression

$$\frac{dw}{dr} = \frac{2A_0 r \ln r + 2C_0 r}{r} + \frac{dw_p}{dr}$$

B_0 must be zero. The shear force

$$Q_r = -D \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} \left(r \frac{dw}{dr} \right) \right) = -\frac{p_0 r}{a} \quad (13)$$

must also be finite, so $A_0 = 0$. The deflection simplifies to

$$w = C_0 r^2 + D_0 + w_p \quad (14)$$

To obtain the particular solution, integrate the operator from Eq.(11) with $p(r) = p_0 r/a$ four times:

$$\begin{aligned} \nabla^4 w &= \frac{p_0 r}{aD} \\ \nabla^3 w &= \frac{p_0}{aD} \\ \nabla^2 w &= \frac{p_0 r}{aD} \\ \nabla w &= \frac{p_0 r^2}{2aD} \\ w &= \frac{p_0 r^3}{6aD} \end{aligned} \quad (15)$$

Note that the factor of a in the denominator is essential for dimensional consistency: $p(r)/D = p_0 r/(aD)$ carries a throughout the integration. Substituting Eq.(15) into Eq.(14), the total deflection is

$$w = C_0 r^2 + D_0 + \frac{p_0 r^3}{6aD} \quad (16)$$

$$w = C_0 r^2 + D_0 + \frac{p_0 r^3}{6aD} \quad (17)$$

The radial bending moment is

$$M_r = -D \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} \left(r \frac{dw}{dr} \right) \right) = -\frac{p_0 r^2}{2a} \quad (18)$$

Part a: Simply Supported

$$@ r = a \rightarrow w(a) = 0, \quad M_r(a) = 0 \quad (19)$$

Applying $M_r(a) = 0$ to Eq.(18):

$$C_0 = c_3 = -\frac{p_0 a^2}{90D} \cdot \frac{4 + \nu}{1 + \nu}$$

Applying $w(a) = 0$ to Eq.(16):

$$C_0 = c_3 = -\frac{p_0 a^2}{90D} \cdot \frac{4 + \nu}{1 + \nu}$$

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$$C_0 = c_3 = -\frac{p_0 a^2}{90D} \cdot \frac{4 + \nu}{1 + \nu} \quad (20)$$

Substituting into Eq.(16):

$$C_0 = c_3 = -\frac{p_0 a^2}{90D} \cdot \frac{4 + \nu}{1 + \nu} \quad (21)$$

The maximum deflection occurs at $r = 0$:

$$C_0 = c_3 = -\frac{p_0 a^2}{90D} \cdot \frac{4 + \nu}{1 + \nu} \quad (22)$$

Part b: Clamped

$$@ r = a \rightarrow w(a) = 0, \quad \left. \frac{dw}{dr} \right|_{r=a} = 0 \quad (23)$$

Applying $dw/dr = 0$ at $r = a$ to Eq.(17):

$$C_0 = -\frac{p_0 a^2}{90D}$$

Applying $w(a) = 0$ to Eq.(16):

$$C_0 = -\frac{p_0 a^2}{90D}$$

$$C_0 = -\frac{p_0 a^2}{90D}$$

The complete deflection for the clamped plate is therefore:

$$C_0 = -\frac{p_0 a^2}{90D} \quad (24)$$

$$w_{\max,cl} = w_{cl}(0) = \frac{p_0 a^4}{150D}$$

For $\nu = 1/3$, comparing $w_{\max,ss}$ and $w_{\max,cl}$:

$$\frac{w_{\max,ss}}{w_{\max,cl}} = \frac{1}{150} \left(\frac{16}{5} + \frac{16}{5} \right) = \frac{32}{150} = \frac{16}{75}$$

Part c: Radial Stress Comparison

The radial stress at the plate surface ($z = t/2$) is

$$\sigma_r = \frac{6M_r}{t^2} \quad (25)$$

Substituting the constants from Parts a and b into Eq.(18):

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{16}{5} + \frac{16}{5} \right) \quad (26)$$

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (27)$$

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (28)$$

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (29)$$

Evaluating at key locations:

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (30)$$

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (31)$$

Taking the difference of Eqs.(28) and (29):

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (32)$$

The two radial stress distributions differ by a constant, independent of r .

Part d: Normalized Non-Dimensional Figure

Using Eq.(17), the tangential moment is

$$M_r = \frac{p_0 a^4}{150D} \left(\frac{16}{5} + \frac{16}{5} \right) \quad (33)$$

The term $-2C_0D(1 + \nu)$ is identical to that in M_r (Eq.(18)). Substituting the constants: For simply supported, using $-2C_3D(1 + \nu) = p_0 a^2(4 + \nu)/45$:

$$\sigma_r = \frac{6}{t^2} \left(\frac{p_0 a^4}{150D} \right) \left(\frac{32}{5} \right) \quad (34)$$

$$\text{[Redacted Equation]}$$
 (32)

For clamped, using $-2C_0D(1 + \nu) = p_0a^2(1 + \nu)/45$:

$$\text{[Redacted Equation]}$$
 (33)

Defining $\rho = r/a$ and $\sigma_0 = p_0(a/t)^2$, the four normalized stresses are

$$\text{[Redacted Equation]}$$
 (34)

$$\text{[Redacted Equation]}$$
 (35)

$$\text{[Redacted Equation]}$$
 (36)

$$\text{[Redacted Equation]}$$
 (37)

The normalized constant difference is

$$\text{[Redacted Equation]}$$
 (38)

Since Eq.(38) is independent of ρ and ν , a normalized non-dimensional figure exactly analogous to Figure 4.4 of the notes is possible. Only the two clamped curves (σ_r and σ_θ) need to be drawn; reading them against the upper base line gives the clamped stresses, and reading the same curves against the lower base line (shifted down by $2\sigma_0/5$) gives the simply supported stresses — identical in construction to Figure 4.4.

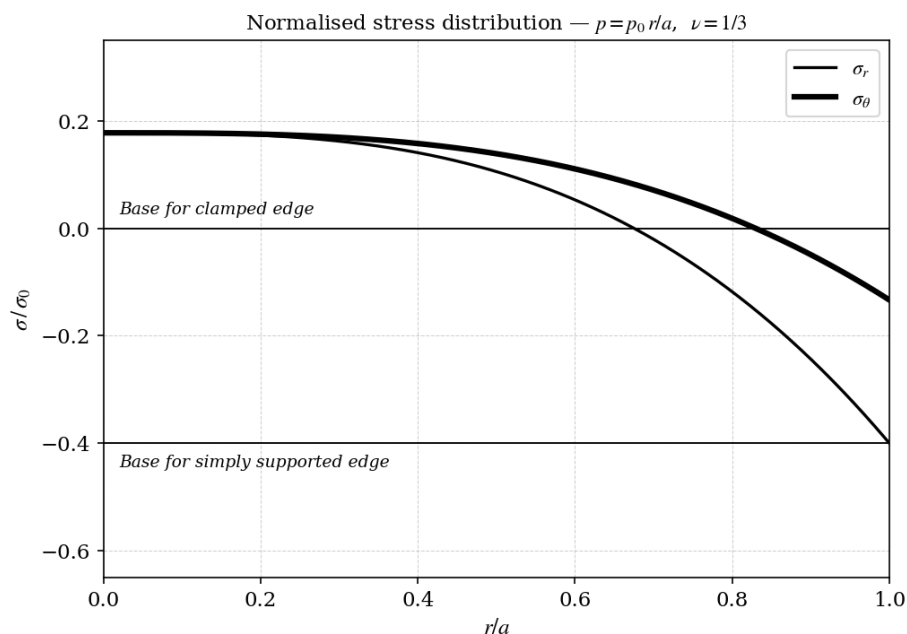


Figure 2: Normalised stress distribution at the bottom surface ($z = t/2$) of a solid circular plate under $p = p_0 r/a$, $\nu = 1/3$.

Q3: A pump diaphragm can be approximated as an annular plate under a uniformly distributed surface load P_0 and with outer edge simply supported (Case 5 in Table 4.3). Compute, using the method of superposition, the maximum plate deflection for $b = a/4$ and $\nu = 0.3$. Compare the result with that given in the table.

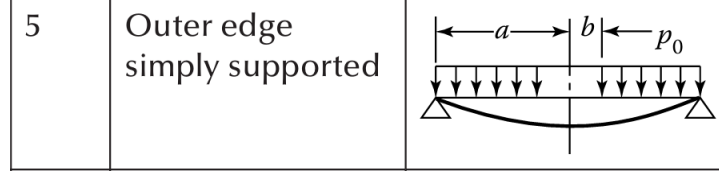


Figure 3: Case 5 in Table 4.3 from Ugural's book

Solution

According to Table 4.3, the loading is as shown in the following figures. For the superposition method, we decompose the problem into three parts:

1. A solid circular plate under the full uniform surface load P_0 , simply supported at the outer edge ($r = a$).
2. An annular plate with no surface load, simply supported at $r = a$, subjected only to an inner edge line force $Q_b = -Q_{r,A}(b)$ at $r = b$.
3. An annular plate with no surface load, simply supported at $r = a$, subjected only to an inner edge line moment $M_b = -M_{r,A}(b)$ at $r = b$.

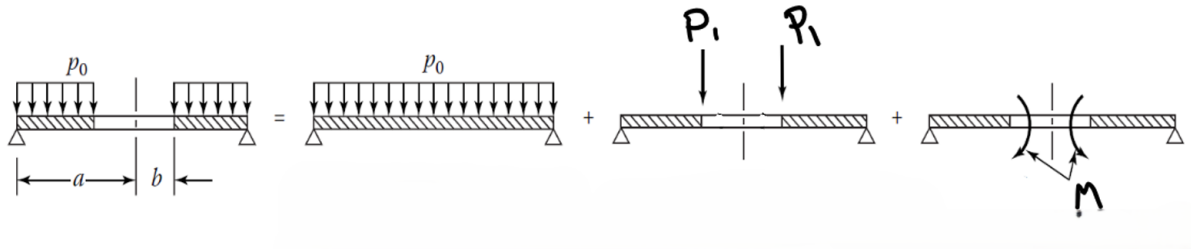


Figure 4: Annular plate 3 positions

The total deflection is the sum of all three parts:

$$w(r) = w_1(r) + w_2(r) + w_3(r) \quad (39)$$

The general deflection is the sum of a homogeneous solution w_h and a particular solution w_p :

$$w_h(r) = A_0 r^2 \ln r + B_0 \ln r + C_0 r^2 + D_0 \quad (40)$$

The internal radial bending moment $M_r(r)$ and radial shear force $Q_r(r)$ are

$$M_r(r) = -D \left(\frac{d^2 w}{dr^2} + \frac{\nu}{r} \frac{dw}{dr} \right) \quad (41)$$

$$Q_r(r) = -D \frac{d}{dr} \left(\frac{d^2 w}{dr^2} + \frac{1}{r} \frac{dw}{dr} \right) \quad (42)$$

Substituting Eq.(40) into Eqs.(41)–(42):

$$\text{[Redacted Equation]}$$

$$\text{[Redacted Equation]}$$

giving the general homogeneous internal forces

$$\text{[Redacted Equation]} \quad (43)$$

$$\text{[Redacted Equation]} \quad (44)$$

Part 1: Solid Circular Plate under Full Uniform Surface Load P_0

For a uniform load P_0 , the particular solution is $w_p(r) = P_0 r^4 / (64D)$. Since the plate is solid, singular terms must vanish: $A_1 = 0$, $B_1 = 0$. The total deflection for Part 1 is

$$\text{[Redacted Equation]} \quad (45)$$

$$\text{[Redacted Equation]} \quad (46)$$

Applying the simply supported boundary conditions:

$$@ r = a \quad \rightarrow \quad w_1(a) = 0, \quad M_{r,1}(a) = 0$$

$$\text{[Redacted Equation]}$$

$$\text{[Redacted Equation]} \quad (47)$$

At $r = b$, Part 1 (Eq.(45)) generates the internal forces cancelled by Parts 2 and 3. From Eqs.(47) and (46):

$$\text{[Redacted Equation]} \quad (48)$$

$$\text{[Redacted Equation]} \quad (49)$$

Part 2: Annular Plate Subjected Only to Inner Edge Line Force Q_b

With no surface load, the solution is purely homogeneous as in Eq.(40). The inner boundary condition cancels the shear force from Eq.(49):

$$@ r = b \quad \rightarrow \quad Q_{r,2}(b) = Q_b = -Q_{r,A} = \frac{P_0 b}{2}$$

From Eq.(44):

$$\text{[Redacted Equation]}$$

The remaining boundary conditions are:

$$@r = b : M_{r,2}(b) = 0 \quad @r = a : M_{r,2}(a) = 0, \quad w_2(a) = 0$$

Writing $M_{r,2}(b) = 0$ and $M_{r,2}(a) = 0$ from Eq.(43) and subtracting to eliminate C_2 :

$$\text{[Redacted Equation]} \quad (50)$$

Substituting Eq.(50) back into $M_{r,2}(a) = 0$:

$$\text{[Redacted Equation]} \quad (51)$$

From $w_2(a) = 0$ using Eq.(51):

$$\text{[Redacted Equation]} \quad (52)$$

Evaluating $w_2(r)$ at $r = b$ using Eqs.(50)–(52):

$$\text{[Redacted Equation]} \quad (53)$$

Part 3: Annular Plate Subjected Only to Inner Edge Line Moment M_b

The inner boundary condition cancels the bending moment from Eq.(48):

$$@r = b \rightarrow M_{r,3}(b) = M_b = -M_{r,A} = -\frac{P_0(3+\nu)}{16}(a^2 - b^2)$$

Since $Q_{r,3}(b) = 0$, from Eq.(44) we get $A_3 = 0$, and the deflection simplifies to

$$\text{[Redacted Equation]} \quad (54)$$

The moment expression from Eq.(41) becomes

$$\text{[Redacted Equation]} \quad (55)$$

Applying the boundary conditions at $r = a$ to Eqs.(54)–(55):

$$\text{[Redacted Equation]}$$

Solving together with the inner BC:

$$\text{[Redacted Equation]} \quad (56)$$

[illegible][illegible]

Summing Eqs.(47), (53), and (58) at $r = b$ per Eq.(39):

$$w_{\max} = w_1(b) + w_2(b) + w_3(b) \quad (59)$$

Substituting $b = a/4$, $\nu = 0.3$, and $\ln(a/b) = \ln 4$ into Eq.(60):

$$w_{\max} = 0.07597 \frac{P_0 a^4}{D}$$

Substituting $D = Et^3/[12(1 - \nu^2)]$ to match the table format $h_5 P_0 a^4/(Et^3)$:

$h_5 = 0.8296$

which agrees with the value given in Case 5 of Table 4.3 ($4 = a/b$).

Table 4.4 Coefficients k and h for Cases 1 through 10 in Table 4.3

a/b	1.25	1.50	2.00	3.00	4.00	5.00
k_1	0.227	0.428	0.753	1.205	1.514	1.745
h_1	0.0051	0.0249	0.0877	0.209	0.293	0.350
k_2	0.135	0.410	1.04	2.15	2.99	3.69
h_2	0.0023	0.0183	0.0938	0.2925	0.448	0.564
k_3	1.10	1.26	1.48	1.88	2.17	2.34
h_3	0.341	0.519	0.672	0.734	0.724	0.704
k_4	0.66	1.19	2.04	3.34	4.30	5.10
h_4	0.202	0.491	0.902	1.220	1.300	1.310
k_5	0.592	0.976	1.440	1.880	2.080	2.19
h_5	0.1841	0.4139	0.6640	0.8237	0.8296	0.813
k_6	0.122	0.336	0.74	1.21	1.45	1.59
h_6	0.0034	0.0313	0.1250	0.291	0.417	0.492
k_7	0.115	0.220	0.405	0.703	0.933	1.13
h_7	0.0013	0.0064	0.0237	0.0619	0.0923	0.114
k_8	0.090	0.273	0.71	1.54	2.23	2.80
h_8	0.0008	0.0062	0.0329	0.1096	0.1792	0.2338
k_9	0.194	0.320	0.454	0.673	1.021	1.305
h_9	0.00504	0.0242	0.0810	0.172	0.217	0.288
k_{10}	0.105	0.259	0.480	0.657	0.710	0.730
h_{10}	0.00199	0.0139	0.0575	0.130	0.162	0.175

Source: Data rearranged from Wahl and Lobo [10].

Figure 5: Table 4.3 from Ugural's book